# Zafar Ahmad

Cardiovascular signal processing · Electrical engineering

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Google Scholar: D5W9TVwAAAAJ · ESC 365: esc365.escardio.org/person/1388860 · Portfolio: zafar-ahmad-5359.github.io/research.html

## RESEARCH PROFILE

Electrical engineer (MS, NUST) studying cardiac electrical activity and its acoustic and pulse consequences through electrocardiograms (ECG), phonocardiograms or heart sounds (PCG) and photoplethysmograms or optical pulses (PPG). I work on the measurement side of cardiac deep learning: the cardiac timing recordings can supply, and how far it can be trusted. My thesis asked whether a heart-sound recording can mark its own cardiac phases; a model that worked on one dataset did not transfer to new hardware, and that result set my direction. In a PhD I want to study cardiac-timing models under distribution shift: how a model can tell that a recording, device or population lies outside validated conditions; how much timing error each clinical reading tolerates; and how models stay calibrated as devices and cohorts change. I bring field acquisition, patient-grouped and pre-registered evaluation and cross-device testing, and want to learn continual learning and physiological modelling.

## EDUCATION

### MS Electrical Engineering — NUST, SEECS

Islamabad, Pakistan | September 2023 – May 2026

Major: Control Systems. CGPA: 3.20/4.00. Thesis grade: A. Degree conferred 18 May 2026.

**Thesis:** Cross-Modal Reconstruction of Cardiac Signals: Decoding ECG-PCG Electromechanical Coupling Dynamics.

**Supervisor:** Nazia Perwaiz; **co-supervisor:** Muhammad Salman Khan.

### BS Electrical Engineering (Communication) — UET Peshawar

Peshawar, Pakistan | September 2018 – November 2022

CGPA: 3.21/4.00. Final-year project grade: A. Degree conferred 3 November 2022.

**Thesis:** Automatic Detection of Paediatric Congenital Heart Diseases from Phonocardiogram Signals.

**Supervisors:** Syed Waqar Shah; Muhammad Salman Khan.

## RESEARCH INTERESTS

Cardiac-phase timing from single-sensor recordings; reliability under distribution shift (patient-grouped and cross-device evaluation, calibration across devices and populations); signal-quality gates for downstream models; wearable ECG–PPG timing; paediatric murmur characterisation.

## TECHNICAL SKILLS

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| **Programming** | Python, MATLAB; PyTorch, TensorFlow/Keras; SciPy, librosa, PyWavelets. |
| **Signal analysis** | STFT/CWT, spectral density, MFCCs, envelopes, peak detection, resampling and artefact handling. |
| **Modelling** | Conditional GANs, attention models, CNNs; copulas and coupled hidden Markov models (ongoing research). |
| **Evaluation** | Patient-grouped cross-validation, pre-registered analysis plans, ablations, calibration, uncertainty, subgroup and cross-device analysis. |
| **Research tools** | Git/GitHub, Linux, GPU training, LaTeX; ECG/PPG/IMU sensor integration. |

## RESEARCH EXPERIENCE

### Research Assistant and Team Supervisor

CAREON · NUST, SEECS / SMV-Lab SINES | December 2025 – present

Continuous AI-Remote E-Health Operations Network. Principal investigator: Nazia Perwaiz.

- Lead technical work on a wearable prototype that records ECG and PPG on a shared clock with inertial sensing for motion-artefact analysis; timing fidelity is the next test.
- Supervise three research apprentices across embedded firmware and sensor integration, signal processing and the cloud data pipeline.
- Set the sampling/acquisition plan and specified an external analogue-to-digital converter (ADC) fallback for precision constraints; develop ECG–PPG timing features and coordinate consenting-volunteer recordings.
- Led a blind cross-device audit of the PCG-to-ECG reconstruction model on 338 record pairs recorded with a different stethoscope at a different chest site; agreement did not transfer without retraining, and the result was reported as found in the JBHI revision.

### Research Assistant (Remote)

Qatar University, College of Engineering | May 2023 – present

Multimodal Cardiac Signal Modelling. Supervisor: Muhammad Salman Khan.

- Developed band-selective augmentation and MFCC-based 1D CNN classification for cardiac recordings, published at EUSIPCO 2024 as first author.
- Develop copula-based coupled hidden Markov models linking ECG waves and heart-sound phases; investigate their dependence across recording conditions.
- Analyse synchronous ECG and PCG using spectral, correlation and rank-based methods; explore noise augmentation that preserves physiological timing.

### Research Apprentice

NCAI, AI in Healthcare Lab, UET Peshawar | June 2021 – May 2022

Supervisor: Muhammad Salman Khan.

- Worked with cardiologists on paediatric heart-sound recording at Rehman Medical Institute, Lady Reading Hospital and six schools, with echocardiography-confirmed annotation.
- Developed collection procedures, consent documents and clinical questionnaires for the institutional review workflow; participated in school screening and referral.
- Built preprocessing, pitch-shift augmentation and 1D CNN models for paediatric congenital heart-disease detection, published as the first-author ICAI 2023 paper.

## CLINICAL AND EMBEDDED RESEARCH

Electronic-stethoscope field recording; clinical collaboration and annotated cardiac datasets; collection protocols and consent workflows; ESP32, AD8232 ECG, MAX30102 PPG and MPU6050 IMU integration. Planned prototype tests cover timing fidelity, acquisition limits and sensitivity to recording hardware.

## SUPERVISION, PROPOSALS AND RECOGNITION

Technical supervision of a junior colleague's 12-lead ECG concept-bottleneck study, presented on his behalf at ESC Congress 2026. Co-wrote two grant proposals submitted in September 2025 (HEC Pakistan NRPU; Qatar University ORS); not awarded.

**Best Summer Intern Award** — NCAI AI in Healthcare Lab, UET Peshawar, 2022. English (IELTS Academic overall 6.5, January 2023); Pashto and Urdu (native).

## PUBLISHED PEER-REVIEWED CONFERENCE PAPERS

**Zafar Ahmad**, Muhammad Salman Khan, Muhammad E. H. Chowdhury, Susu Zughaier, and Wanis Hamad Ibrahim. "Phonocardiogram Classification Based on 1D CNN with Pitch-Shifting and Signal Uniformity Techniques." *32nd European Signal Processing Conference (EUSIPCO)*, Lyon, France, 2024, pp. 1536–1540. https://doi.org/10.23919/EUSIPCO63174.2024.10715037.

**Zafar Ahmad**, Muhammad Zeeshan, Arshad Sohail, Fazal-e-Haq, Muhammad Haris, Misha Urooj Khan, Sayed Shahid Hussain, and Muhammad Salman Khan. "Automatic Detection of Paediatric Congenital Heart Diseases from Phonocardiogram Signals." *3rd International Conference on Artificial Intelligence (ICAI)*, Islamabad, Pakistan, 2023, pp. 188–195. https://doi.org/10.1109/ICAI58407.2023.10136668.

## JOURNAL MANUSCRIPTS IN REVISION (NOT ACCEPTED OR PUBLISHED)

**Minor revision · IEEE Journal of Biomedical and Health Informatics**

**Zafar Ahmad**, Nazia Perwaiz, Muhammad Salman Khan, Sidra Sultana, Fatima Abdullah, Hamza Anjum, and Izharul Haq. "PCG-to-ECG Signal Reconstruction Using Conditional GAN with Multi-Scale Attention." Submitted March 2026; minor revision requested and revised manuscript resubmitted July 2026; second minor revision requested 18 October 2026; revision prepared. Not accepted or published.

**Major revision · European Heart Journal – Digital Health**

**Zafar Ahmad**, Nazia Perwaiz, Muhammad Salman Khan, Suleiman Alabdallah, Sidra Sultana, and Izhar ul Haq. "Synthetic-ECG-gated phonocardiography for paediatric heart-murmur characterisation: a cross-modal deep-learning method." Submitted August 2026; major revision requested 23 September 2026; revised manuscript with a pre-registered re-analysis (analysis plan frozen before any arm was scored), under a revised title, prepared October 2026. Not accepted or published.

## ADDITIONAL MANUSCRIPT UNDER REVIEW

Muhammad Salman Khan, **Zafar Ahmad**, and Hinna Nayab. "Parallel multi-scale analysis for paediatric heart disease classification from phonocardiograms." *IEEE Journal of Selected Topics in Signal Processing*. Submitted November 2025; last reported as under review (status as of 9 October 2026); not accepted or published.

## ESC CONGRESS 2026 ABSTRACTS AND PRESENTATIONS

Munich, Germany · European Society of Cardiology · abstracts published in *European Heart Journal*, volume 47, Supplement 1, October 2026

**Zafar Ahmad (presenter)**, Nazia Perwaiz, Muhammad Salman Khan, Sidra Sultana, and Izharul Haq. "Synthetic ECG-guided temporal gating improves paediatric murmur grading, timing, and shape classification on the CirCor DigiScope dataset." ePoster, 28 August 2026, session *Digital tools in practice: transforming clinical care*. Eur Heart J. 2026;47(Suppl 1):ehag785.4329. https://doi.org/10.1093/eurheartj/ehag785.4329. ESC 365 record.

Title as published. Its results are superseded by the pre-registered October 2026 comparison and should not be cited as evidence; see the current research finding: zafar-ahmad-5359.github.io/research.html#current-evaluation.

Hamza Anjum, Nazia Perwaiz, Sidra Sultana, Tahira Anwar Lashari, and **Zafar Ahmad (presenter)**. "An explainable calibrated concept bottleneck model with evidence-constrained large language model reporting for multi-label valvular heart disease screening from 12-lead ECGs." ePoster, 31 August 2026, session *Artificial intelligence-powered assessment of cardiac function*. Role: technical supervision of the first author and congress presentation. Eur Heart J. 2026;47(Suppl 1):ehag785.4378. https://doi.org/10.1093/eurheartj/ehag785.4378. ESC 365 record.

Manuscript statuses checked 31 October 2026 (JSTSP last reported 9 October 2026). Congress titles reproduce the published abstracts.